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Toward sustainable web design: Greenscan-AICO₂R for AI-based carbon footprint optimization of websites

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Introduction: The rapid growth of internet usage has significantly increased the energy demand of digital infrastructure, contributing to rising global carbon emissions. Modern websites, often characterized by large media assets and complex front-end structures, are major contributors to this environmental burden.

Methods: To address this challenge, proposed Greenscan-AICO₂R (CO₂ Emission Reduction by AI), a practical framework designed to evaluate and enhance the sustainability of websites. Unlike widely used tools such as Website Carbon Calculator or Page Speed Insights, which primarily report performance metrics or provide generic suggestions, proposed framework performs multi-page crawling, applies rule-based image classification to distinguish essential from decorative content, estimates monthly carbon emissions based on page weight and assumed visitor tra_c, and leverages natural language processing (NLP) to generate tailored, content-specific optimization recommendations.

Results: Simulation shows that proposed optimization strategies such as image compression, lazy loading, and optimized media format can help to reduce the web page size as well as carbon emissions by approximately 30%–40%.

Discussion: A representative case study has revealed that the proposed framework helped in achieving a 32% reduction in monthly CO₂ emissions in comparison with the baseline scenario.

Future Direction: Future enhancements include replacing rule-based classification with convolutional neural networks (CNNs), deepening NLP for more context-aware suggestions, supporting JavaScript-rendered content, integrating real-time carbon intensity data, and developing automated resource optimization. By integrating analysis, simulation, and actionable recommendations, Greenscan-AICO₂R empowers web developers to create more energy-efficient and environmentally responsible digital applications.

KEYWORDS

carbon emission estimation, digital carbon footprint, green AI, image classification, natural language processing, sustainable web design, web optimization, website sustainability

1 Introduction

The internet has become integral to modern communication, education, work, and daily activities. However, its rapid expansion has dramatically increased electricity consumption across digital infrastructure, including servers, data centers, network equipment, and end-user devices. A substantial portion of this energy still derives from

fossil fuel sources, making the digital sector a meaningful contributor to global greenhouse gas emissions.

The role of Information and Communication Technologies (ICT) in achieving the Sustainable Development Goals (SDGs) is acknowledged in the United Nations 2030 Agenda for Sustainable Development (United Nations General Assembly, 2015). The agenda recognizes the role of ICT in ensuring access to education and health services, as well as encouraging innovation and resource efficiency. It also recognizes the need to mitigate the harmful effects of ICT on the environment. Digital technologies can contribute to increased resource consumption due to inefficiencies in their use and design. Consequently, the energy efficiency of websites and digital services are acted as the current key research topic in sustainable computing and green software engineering (SE), supporting the main objectives of reducing carbon emissions and achieving the United Nations Sustainable Development Goals (SDGs).

1.1 Research gap

Existing website sustainability assessment tools, such as Website Carbon Calculator, EcoIndex, and PageSpeed Insights, primarily focus on page-level analysis and provide generic performance or carbon emission metrics (Wholegrain Digital, 2025c; Burt, 2023; Cogworks, 2024). While these tools raise awareness of website-related carbon emissions, they generally do not support comprehensive multi-page analysis, distinguish between essential and decorative media resources, or provide context-aware optimization strategies tailored to the structure and content of individual websites (Cogworks, 2024; Wholegrain Digital, 2025c). Furthermore, most existing approaches estimate carbon emissions on a per-pageview basis without accounting for overall website structure, visitor traffic, and cumulative environmental impact, limiting their practical applicability for real-world website sustainability assessment (Coroama et al., 2014; Wholegrain Digital, 2025c; The Green Web Foundation, 2024).

The proposed research is to design and evaluate a practical, integrated framework as Greenscan-AICO₂R that moves website carbon assessment beyond the surface-level, single-page metrics offered by existing tools. Specifically, the study aims to (i) enable automated, multi-page analysis of website structure and media content rather than isolated single-page snapshots; (ii) distinguish essential from decorative visual content through heuristic image classification, so that optimization efforts can be targeted where they yield the greatest environmental benefit; (iii) estimate monthly carbon emissions at the website level using page weight and projected visitor traffic, providing a more realistic measure of real-world impact than per-pageview estimates alone; (iv) generate context-aware, actionable optimization recommendations through natural language processing rather than generic, one-size-fits-all suggestions; and (v) simulate the emission and performance impact of these recommendations before implementation, giving developers evidence-based guidance rather than abstract advice. This study seeks to answer the following research questions:

1. How can AI be used to measure the carbon footprint of websites effectively?
2. What factors contribute most to the energy consumption of websites?
3. How can AI-based optimization techniques reduce website-related carbon emissions?
4. What is the impact of sustainability optimization on website performance and user experience?

The rest of this paper is structured as follows. In Section 2, background literature is provided, along with a review of related works that deal with the issues of digital pollution, carbon footprinting on the website level, and the available carbon assessment tools, positioning Greenscan-AICO₂R in the context of existing literature on sustainable web design. Section 3 contains information on the materials and methods employed, including a description of the system workflow, the framework architecture, data gathering and implementation process, carbon emission estimation algorithm, and computational complexity analysis. In Section 4, Results deals with the assessment of optimized carbon emissions on the websites of various types, government and private sectors included. Section 5 (Discussion) explains the analysis of the results, compares the proposed methodology with existing tools, and discusses the limitations and validity of the proposed framework.

2 Background

2.1 Digital pollution and environmental impact

Digital pollution is the environmental effects of our day-to-day activities involving digital technologies. The effects of digital pollution arise as a consequence of the proliferation of digital infrastructure, which includes data centers, communication infrastructure, and always-connected devices. The environmental effects of digital products and services can be measured in terms of the greenhouse effect, which is typically measured in metric tons of CO₂ equivalent, factoring in the cumulative effect of the release of different gases into the atmosphere (Burt, 2023; Lannelongue et al., 2021).

The concept of digital pollution encompasses multiple dimensions, such as e-waste generation, energy consumption during operation, and pollution from the manufacturing and disposal of digital devices (e.g., computers, smartphones, televisions, and appliances). Energy consumption analysis is influenced by numerous factors, including hardware efficiency, the number of internet users and services, and technological advancements that improve overall performance (Cogworks, 2024).

Growing concerns about the environmental footprint of digital technologies have attracted significant research attention in recent years (Coroama et al., 2014; Yadav et al., 2024; Vinuesa et al., 2020; Gohr et al., 2025). Early studies primarily examined energy demands of large-scale data centers and networks. For instance, Belkhir and Elmeligi (2018) highlighted the expanding carbon footprint of information and communication technologies (ICT),

warning that unchecked growth could elevate the sector's energy demand to levels comparable with major industries. They stressed the urgency of hardware and software optimizations.

More recent efforts have attempted to integrate crawling, classification, and recommendation into a single system (Root Web Design Studio, 2025; Cogworks, 2024; Clearcut Derby, 2025). Related work has also explored automated systems that scan websites and modify code to minimize energy use, though such approaches often lack a mechanism to explain their recommendations to developers. Recent work has also been done on natural language processing (NLP) for website analysis (Cogworks, 2024; Google, 2025). Calculation methodologies for carbon footprint have evolved from simplistic approaches such as fixed multipliers based on page sizes to more advanced approaches considering real-world factors such as device and traffic patterns (Wholegrain Digital, 2025a,c; The Green Web Foundation, 2024; Coroama et al., 2014). Although, these developments improve the accuracy of predictions, their complexity might make their integration into development processes difficult.

A critical consideration in sustainability research is balancing environmental benefits with user experience, with emission reductions expected not to compromise usability (Todorov and Stefanova, 2025; Mårtensson, 2023; Clearcut Derby, 2025; Henderson, 2020). Nevertheless, prevailing approaches tend to focus narrowly on emission estimation, scanning, or speed optimization, often neglecting holistic trade-offs.

Widely available online tools such as EcoIndex and WebsiteCarbon provide quick per-page carbon estimates (Wholegrain Digital, 2025c,a; Cogworks, 2024; Burt, 2023; Sustainable Web Design, 2025; Compare Your Footprint, 2024; The Public Good, 2021). While effective in raising awareness, they lack multi-page analysis, intelligent media differentiation, and site-specific optimization guidance (Wholegrain Digital, 2025b), limiting their utility for comprehensive sustainability audits.

The proposed Greenscan-AICO₂R framework directly addresses these limitations by integrating automated crawling, heuristic image classification, carbon footprint estimation, and NLP-driven recommendations into a cohesive tool (Yadav et al., 2024; Vinuesa et al., 2020; Gohr et al., 2025). It delivers in-depth, interpretable sustainability insights without requiring pre-labeled datasets, adapting outputs to each website's unique structure and resources, and presenting results through graphical and textual reports.

3 Materials and methods

This research work conducts and evaluates the energy consumption and greenhouse carbon emissions associated with private, government websites. The research centers on estimating the carbon footprint of websites using webpage resource characteristics, page size, projected visitor traffic, and a simplified carbon estimation model. The proposed Greenscan-AICO₂R framework serves as a diagnostic tool to assess efforts toward reducing the digital environmental footprint across diverse website types and to evaluate alignment with Sustainable Development Goals (SDGs) related to digital sustainability. The methodology combines a comprehensive literature review with empirical

data analysis from real-world websites. Various categories of websites were taken into consideration, such as Amazon, Flipkart, Wikipedia, university portals, banking websites, news websites, and social media platforms. For the purpose of this investigation, 1,000 government and private university websites were chosen, which included government and private university websites with different structures and images. The calculation of carbon emissions was carried out based on the structure of the websites.

3.1 Overview of proposed Greenscan-AICO₂R wWorkflow

The proposed Greenscan-AI CO₂R workflow is shown in Figure 1.

- **Input:** The system starts by taking the website URL as an input from the user and then proceeds with the verification of the website's accessibility.
- **Processing:** The next step is to crawl the website pages and retrieve the internal links and image URLs. Then, the images are filtered as necessary or not necessary based on the proposed rule-based heuristic. The carbon emission for a month is also estimated by considering the total media size and the traffic. The HTML content is processed by the NLP engine to provide the optimization recommendations for the website pages.
- **Output:** A visual report is generated to display the results.

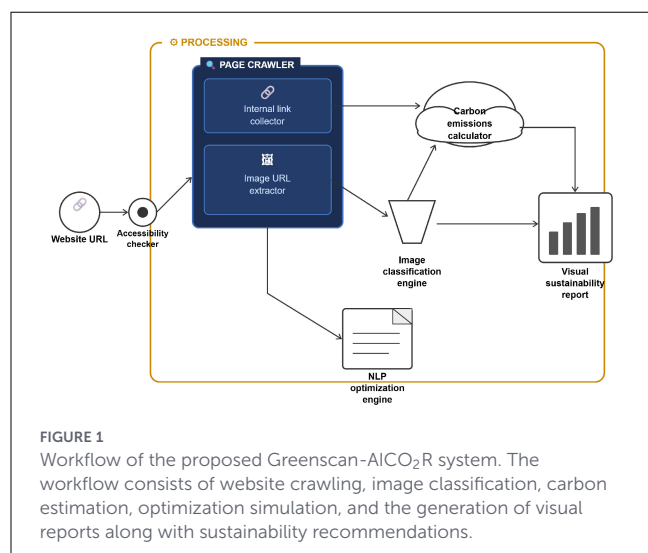
The overall framework system provides a complete end-to-end solution for website sustainability assessment.

3.2 Detailed Greenscan-AICO₂R framework

Greenscan-AICO₂R is a website sustainability assessment framework that focuses on the energy consumption and environmental impact of websites. Unlike traditional auditing tools that analyze a webpage sequentially, the proposed framework is an integrated approach which includes web crawling, image analysis, calculation of carbon footprint, and Natural Language Processing (NLP)-based recommendations provides an auditing approach that provides specific metrics and an explanation of those metrics in a simple and visual manner.

The framework's core components include:

1. A multi-page web crawler that is used to extract internal links, images, and other key resources.
2. A heuristic classifier that separates essential images from purely decorative ones, based on content and usage context.
3. A carbon estimation module is used to calculate the monthly CO₂ emissions from page size and projected traffic.
4. An NLP recommendation engine that finds optimization opportunities such as missing metadata, inefficient loading, and redundant assets.
5. A reporting module that compiles all findings into a structured Portable Document Format(PDF) document with visualizations.



3.3 Data collection and implementation

The framework was tested with various websites with different designs, structures, and media usage. The key aim was to test the accuracy of the results obtained in data extraction, image classification, emission estimation, and recommendation. Greenscan-AICO₂R was written in Python 3.10 (Python Software Foundation, Wilmington, DE, USA; <https://www.python.org/>) with libraries like requests for handling HTTP requests, BeautifulSoup for crawling websites, matplotlib for visualization, and fpdf for generating PDF reports. The results showed an average reduction of 32 percentage in estimated emissions by analyzing 1,000 websites, including government and private websites, consistent with prior empirical findings on the energy costs of unoptimized web content (Chen et al., 2023; Das et al., 2025). The tool works locally and generates PDF reports with charts and graphs. It is represented in (Algorithm 1).

The detailed architecture is shown in Figure 2.

3.3.1 Setup configuration

Greenscan-AICO₂R framework was developed using Python 3.11 in Visual Studio Code on a Windows 11 operating system running on an Intel Core i5 processor with 8 GB RAM. Website crawling and analysis were performed using Requests and BeautifulSoup, while Scikit-learn (TF-IDF and Logistic Regression) supported NLP-based recommendation generation. TensorFlow was included for future deep-learning enhancements. Data visualization was implemented using Matplotlib and Plotly, Streamlit provided the web interface, FPDF generated PDF reports, and TextStat was used for readability analysis. The framework was tested using Google Chrome (Version 137 or later) and Microsoft Edge (Version 137 or later). To evaluate the robustness and generalizability of the framework, websites were selected from multiple domains, including online encyclopedias, government portals, educational institutions, commercial e-commerce

platforms, and news/blog websites. Representative examples included Wikipedia, Digital India, the National Informatics Centre (NIC), Amazon, Flipkart, and selected university websites. These domains were chosen because they exhibit diverse webpage structures, media densities, navigation complexities, and user traffic patterns, enabling evaluation under realistic web environments.

3.3.2 Web crawling and data collection

The crawler was evaluated on websites of varying sizes and structures through multiple execution runs. It successfully extracted HTML content, metadata, image URLs, and hyperlinks from the majority of tested websites. Pages protected by CAPTCHA, login authentication, or restrictive robots.txt policies were skipped and recorded as inaccessible to avoid incomplete or unauthorized data extraction.

3.3.3 Image classification observations

These images were either essential (content-related or functional) or decorative (aesthetic) in nature. On an average, 65%–80% of the images fell into the category of being decorative and about 20%–35% of the images fell into the category of being essential image. Hence, there was a tremendous potential for optimization by reducing the media.

3.3.4 Carbon footprint measurement

Carbon emissions were estimated using a simplified model derived from the Website Carbon methodology (Burt, 2023)

$$\text{CO}_{2\text{monthly}} = \frac{P_s \times V \times f}{1000} \quad (1)$$

where P_s is the average page size (KB), V is the estimated monthly number of visitors (default: 1,000), and f is the carbon emission factor (default: 0.81 g CO₂e/MB), representing the estimated greenhouse gas emissions associated with transferring and processing one megabyte (MB) of web data. The carbon emission factor of 0.81 g CO₂e/MB was adopted from the Website Carbon Calculator methodology proposed by Wholegrain Digital (2025c), as it is a widely used benchmark for estimating the carbon footprint of web data transfer and enables consistent comparison of website sustainability across different studies (Coroama et al., 2014; Wholegrain Digital, 2025c). Optimization Efficiency and Emission reduction are calculated in Equation 2, 3.

Optimization efficiency is calculated as:

$$\eta = \frac{S_0 - S_1}{S_0} \times 100 \quad (2)$$

Emission reduction is calculated as:

$$\Delta E = E_0 - E_1 \quad (3)$$

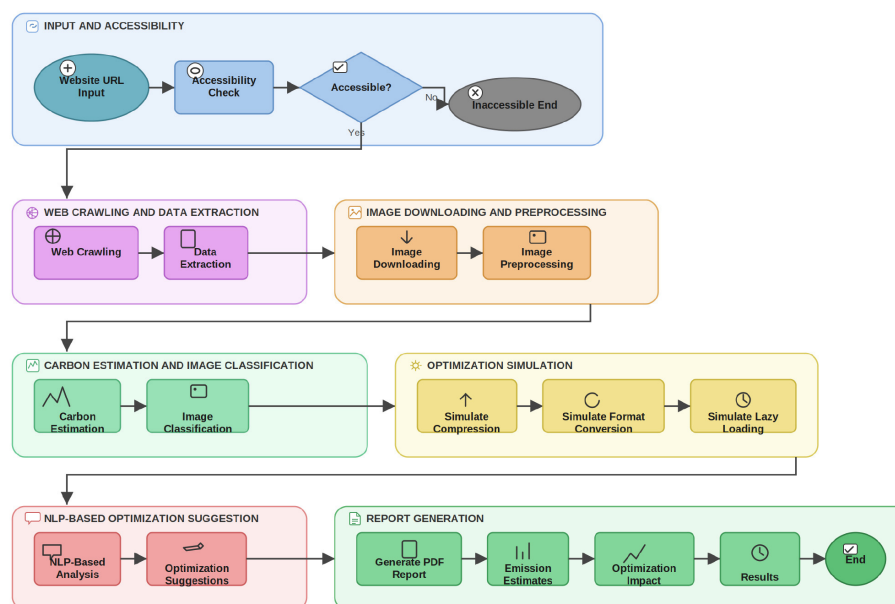


FIGURE 2

Detailed architecture of Greenscan-AICO₂R. The pipeline includes various steps such as accessibility checking, crawling, media classification, emission estimation, optimization simulation, and recommendations.

Larger page sizes showed a strong correlation with higher emissions, reinforcing the importance of media optimization. Consider a webpage with a size of approximately 2,705 KB, was estimated to produce approximately 2,201 kg of CO₂/month. The results highlighted the close relationship between webpage size and carbon emissions. Furthermore, observation of the results shows that the website size increases, the estimated CO₂ emissions also tend to increase. However, absolute carbon emissions can be varied across websites based on design choices, media content, and resource composition, the result indicate that reducing unnecessary webpage resources consistently can minimize the environmental impact of websites, thereby supporting more sustainable web design practices.

3.3.5 NLP-based optimization suggestion

The NLP module employs TF-IDF and similarity scoring for pattern detection, such as identifying “too much JavaScript” or “missing alt text.” Suggestions are made based on rule-based and context-dependent, like “Compress decorative images over 40% of total”.

3.3.6 Suggestion generation

Suggestions are varied for each and every site which shows the presence of various problems such as the absence of meta tags, high script usage, and absence of structural content.

3.3.7 Report generation and output visualization

The output results were shown in the form of visualization via pie charts, bar graphs, and comprehensive PDF reports, which will be better understanding of environmental impact and recommended actions.

3.4 Computational complexity

Computational Complexity calculates with input size:

- Web crawling: $\mathcal{O}(P)$.
- Image classification: $\mathcal{O}(I \cdot f)$, where f acts as inference cost.
- Analysis of NLP: $\mathcal{O}(P \cdot T)$, where T acts as token count.

```

1: Input: Base website URL
2: Extract internal links (max depth = 4)
3: for each page do
4:   Download images and compute total size
5:   Classify images (essential/decorative)
6:   Estimate carbon footprint  $E$ 
7:   Generate optimization suggestions using NLP
8: end for
9: Aggregate total emissions and improvements
10: Generate summary report and visual analytics (PDF + charts)
  
```

Algorithm 1. Multi-page analysis for Greenscan-AICO₂R.

TABLE 1 Summary of reduction statistics.

Metric	Value
Mean reduction (%)	32.0
Standard dDeviation	3.5
95% Confidence interval	± 3.5
Minimum reduction (%)	28.4
Maximum reduction (%)	38.0

Overall calculation of computational complexity shows $\mathcal{O}(P + I \cdot f + P \cdot T)$, linear and very efficient for website of medium-to-large size. Further, extensive library ecosystem of Python supported the implementation of the proposed framework. Many libraries, such as Requests and BeautifulSoup, enable efficient web crawling and HTML parsing, and matplotlib is used to generate visualization, and facilitated automated report creation. The most efficient and adopted library system allowed for rapid development and integration of the framework components to be maintained without requiring computationally intensive infrastructure.

4 Results

The proposed Greenscan-AICO₂R framework was evaluated by collecting data from a large number of websites such as government sector portals, educational institutions, commercial platforms, Wikipedia pages, and news/blog websites. The evaluation examined webpage resource characteristics, estimation of carbon emission, heuristic image classification, optimization simulation, and NLP-based recommendation generation. The following subsections present the experimental findings and discuss their implications for sustainable website design.

4.1 Calculation of optimized carbon emissions for common websites

Table 1 presents a statistical summary of the estimated CO₂ emission reductions across the evaluated websites. To provide basic statistical validation of the optimization results, summary statistics were computed for the observed emission reductions. The results demonstrated relatively consistent optimization performance with limited variation across different website categories.

Table 2 presents the estimated monthly CO₂ emissions before and after applying the recommended optimizations for five representative websites. The results demonstrate consistent reductions, with optimization efficiencies ranging from 36.2% to 38.0%.

4.2 Calculation of optimized carbon emissions for Wikipedia websites

The estimated monthly CO₂ emissions of multiple Wikipedia pages were evaluated before and after applying the optimization strategies proposed in the Greenscan-AICO₂R framework.

After applying the framework's optimization strategies, estimated monthly CO₂ emissions dropped from around 2,200 kg to about 1,500 kg—a reduction of roughly 700 kg per month. That drop is a reasonably strong signal that the targeted optimizations actually work, not just in theory but in measurable terms.

4.3 Comparison overview of carbon emission estimates

Table 3 below offers a summary of the estimated carbon emissions of these popular public tools—Website Carbon, Website Emissions, and pTools—compared to the proposed Greenscan-AICO₂R framework.

While the most existing systems report emissions on a per-page basis (usually in grams of CO₂ per view), Greenscan-AICO₂R takes a more holistic approach to monthly aggregation, considering into account multi-page structures and estimated visitor traffic. The framework reduced the total monthly emissions by around 32% after optimization, demonstrating the benefit of targeted, content-aware interventions to reduce unnecessary data transfer and energy consumption. The media analysis found that only a small number of the images used were critical to the overall content of the page, with the vast majority being purely decorative. This is a critical optimization opportunity for image compression and format conversion to WebP or other optimized formats for faster page load times and reduced carbon footprint impacts. Page weight reduction is estimated at one-third of the original page weight ranges of 40–120 KB. The visualizations provided by the tool—pie charts for image composition, side-by-side image size comparisons, and emission trend charts—proved to be critical for effective analysis and pattern identification such as excessive decorative media or redundant script usage.

4.4 Comparison of original and optimized carbon emissions for government and private websites

Figure 3 compares original and optimized CO₂ emissions for a selection of government websites. A consistent reduction trend was observed, with high-traffic domains (e.g., digitalindia.gov.in) showing the greatest absolute benefits from media and size optimizations.

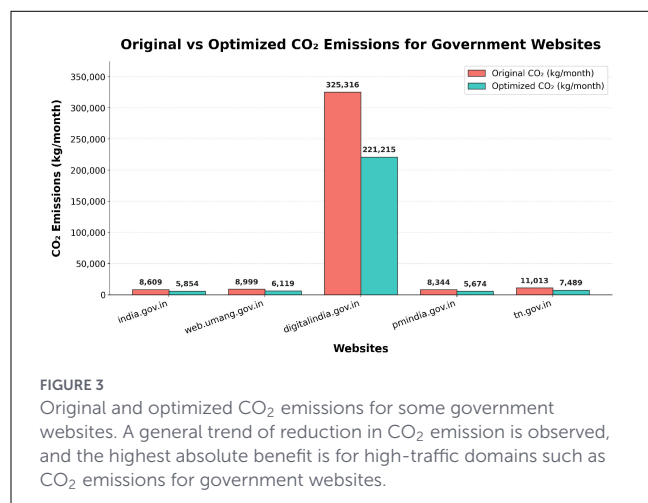
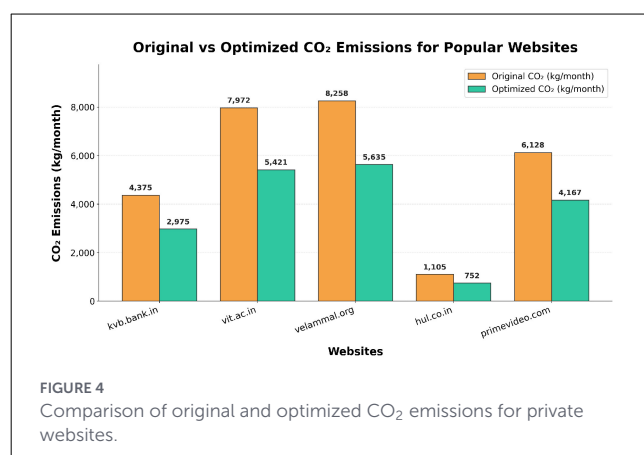
Figure 4 presents similar results for private/commercial and educational websites. Regardless of design complexity, all evaluated sites exhibited clear emission reductions post-optimization, demonstrating the framework's broad applicability.

TABLE 2 Estimated CO₂ emissions, optimized CO₂ emissions, and percentage reduction for selected websites.

Website/category	Data size (MB)	Estimated CO ₂ (kg/month)	Optimized CO ₂ (kg/month)	Reduction (%)
Wikipedia (Information)	52.8	6.2	3.9	37.1
Digital India (Government)	78.5	8.9	5.6	37.0
VIT Chennai (Educational)	41.3	5.1	3.2	37.3
Amazon India (E-commerce)	93.4	10.5	6.7	36.2
The Hindu (News)	60.2	7.1	4.4	38.0

TABLE 3 Comparative estimation of carbon emission between existing tools and Greenscan-AICO₂R.

Tool	Metric Basis	Estimated CO ₂	Rating
Website carbon	kg CO ₂ /pageview	0.000360	F
Website emissions	kg CO ₂ /pageview	0.00259	F
pTools calculator	kg CO ₂ /pageview	0.00505	F
Greenscan-AICO ₂ R (original)	kg CO ₂ /month	10774.72	N/A
Greenscan-AICO ₂ R (optimized)	kg CO ₂ /month	7326.81	Optimized



4.5 Comparison of existing tools vs. proposed system

The available tools have primarily focused on single-page emission estimates, lacking depth. On the other hand, Greenscan-AICO₂R has enhanced capabilities with multi-page crawling, heuristic image classification, natural language processing-based recommendations, optimization simulation, and visual/reporting generation. The comparison of existing tools with proposed system is shown in Table 4.

5 Discussion

The results indicate that content-aware optimization, particularly targeting decorative media, can meaningfully reduce a website's estimated carbon footprint without requiring server-side or infrastructure changes. The consistency of the reduction range (28.4%–38.0%, Table 1) across government, educational, and commercial site categories suggests that the underlying inefficiency, namely unoptimized decorative imagery, is a structural pattern in common web design practice rather than an artifact specific to any one sector.

The comparison against existing calculators (Table 3) also points to a methodological difference worth addressing directly: Website Carbon, Website Emissions, and pTools Calculator report emissions on a per-pageview basis, whereas Greenscan-AICO₂R reports monthly totals that scale with assumed traffic. These two kinds of figures are not directly comparable, and the monthly total depends heavily on the traffic assumption (V in Equation 1) rather than on page design alone. Normalizing Greenscan-AICO₂R's output to a per-1,000-pageviews basis, alongside the monthly total, would allow direct benchmarking against these existing tools and is planned as a near-term refinement (see Section 6.1).

The feature gap identified in Table 4 also reflects a genuine trade-off familiar to sustainability tooling in general: single-page calculators are lightweight and immediate, whereas Greenscan-AICO₂R's multi-page crawling and classification pipeline trades speed for depth. Whether this trade-off is worthwhile depends on the use case, such as a one-off audit versus ongoing monitoring,

TABLE 4 Feature comparison between existing tools and Greenscan-AICO₂R.

Feature	Website carbon	Website emissions	pTools carbon	Greenscan-AICO ₂ R (proposed)
Page-level carbon estimate	✓	✓	✓	✓
Multi-page analysis	✗	✗	✗	✓
AI/NLP-driven recommendations	✗	✗	✗	✓
Image classification (essential vs. decorative)	✗	✗	✗	✓
Optimization simulation and emission comparison	✗	✗	✗	✓
Automated report generation (PDF + charts)	✗	✗	✗	✓

and is a question better answered by future user studies than by this evaluation alone.

A further consideration is the possibility of a rebound effect: as optimization tools make sustainable web design cheaper and easier to adopt, the resulting efficiency gains could, paradoxically, encourage greater overall web traffic or content production, a dynamic consistent with the Jevons Paradox observed in other domains of energy-efficient technology adoption (Fich et al., 2022).

5.1 Limitations

Several limitations should be considered when interpreting these results. First, the framework estimates carbon emissions from page size and an assumed visitor count rather than directly measured energy consumption or a hosting provider's actual energy mix; the resulting figures are therefore best read as relative, comparative estimates rather than absolute measurements. Second, the rule-based heuristic image classifier may misclassify complex or context-dependent imagery, for example inline diagrams that are decorative in one context but essential in another, and JavaScript-rendered content is only partially supported, which could underestimate the footprint of highly dynamic sites. Third, the evaluation set was drawn primarily from government, educational, and commercial websites; while the observed reduction range was consistent across these categories, further validation on a broader, internationally diverse sample is needed before generalizing the findings beyond this context. Finally, the study did not include validation against ground-truth energy measurements, such as server-side power monitoring, which would be needed to confirm the absolute accuracy of the emission estimates rather than their relative, before/after comparison.

5.2 Validity

The proposed Greenscan-AICO₂R framework explained its validity through evaluation across various categories of website such as government, private, commercial, and news/information websites. It is an integrated approach which includes web crawling, image analysis, calculation of carbon footprint, and Natural Language Processing (NLP)-based recommendations provides an

auditing approach that provides specific metrics and an explanation of those metrics in a simple and visual manner. Experimental results showed a consistent reduction of approximately 30%–40% in estimated carbon emissions by applying the recommended optimization strategies. The use of the proposed framework enhances the reliability and consistency of the results. This website sustainability assessment framework focuses on the energy consumption and environmental impact of websites.

6 Conclusion

This study demonstrated that proposed Greenscan-AICO₂R framework uses AI to reduce the carbon footprint of websites. This framework integrated multi-page crawling, heuristic image classification, carbon emission estimation, and NLP-based recommendation generation into a unified, automated workflow of sustainability assessment. Analysis of various websites, such as government and private websites, revealed that webpage size, unoptimized media content, and overall design composition are the dominant contributors to energy consumption and associated CO₂ emissions. The AI-driven optimization techniques of proposed framework, including image compression, lazy loading, optimized media formats, and elimination of redundant resource are achieved to reduce the carbon emission of approximately 30%–40%. The performance of websites were enhanced by reducing size of the page and resource loading which results in faster content delivery and improved user experience. The overall findings showed that AI-based sustainability optimization offers a practical and scalable environment to reduce the environmental impact of web infrastructure. It improved the performance outcomes and valuable tools for future green web development practices.

6.1 Future work

Future enhancements include replacing rule-based classification with convolutional neural networks (CNNs), deepening NLP for more context-aware suggestions, supporting JavaScript-rendered content, integrating real-time carbon intensity data, and developing automated resource optimization. Since deep

learning models carry their own training and inference energy costs (Gupta and Tiwari, 2024), any such enhancement would need to be weighed against its own footprint, though recent efficiency gains in ML training suggest this cost is falling over time (Patterson, 2025).

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

MD: Writing – original draft, Methodology, Conceptualization, Investigation, Resources, Funding acquisition. SM: Visualization, Validation, Methodology, Writing – original draft, Supervision, Project administration, Writing – review & editing. AM: Writing – review and editing, Visualization, Conceptualization. JL: Data curation, Conceptualization, Funding acquisition, Writing – review & editing, Formal analysis.

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